Week 1 Deliverables.

1- Dataset Overview and Documentation

A structure summary that includes the first:

1. **Number of rows** = 7544
2. **Number of columns** = 80
3. **Variable types:**
    - String variables,
    - Date/Time variables,
    - Numerical variables,
    - special cases or mixed variables,
    - Boolean variables
4. **General description of dataset themes**:

The Dataset appears to cover a record of Student applications, carrying their information which covers data related to their academic programme at DePaul University. It primarily captures: Educational background, Demographic and Admission related details for individuals or students who have applied to the University.

2- Data Dictionary (Spreadsheet link)

https://docs.google.com/spreadsheets/d/13pPWJVPGTm7XYv0jMXpvcTy6a32Fz1bDKLWpsKML9aU/edit?usp=sharing

3- Data Quality Report:

Dataset Overview:

- Total Rows: 7,544
- Total Columns: 80
- Duplicate Rows: 1

1- Missing Value Analysis:

Here are some key findings:

Table 1.0: Table showing some Columns variables_names, missing counts and missing percentages.

Column	Missing Count	Missing %
Major	7,544	100%
College	73	0.97%
Given_Name	4	0.05%
Last_Name	2	0.03%

Interpretation:

- Major is completely empty → this column is unusable unless populated.
- Most other fields have very low missing rates, which is a strong sign of good dataset completeness.

Duplicate Record Check:

- Only 1 duplicate row found.

2- **Outlier/Unusual Entries:**

Detected mainly in timestamp fields:

Table 2.0: Table showing some columns variable_name and outlier counts.

Column	Outlier Count
Recieved_At	1,015
Created_At	1,015
Modified_At	1,015

Interpretation:

These are likely:

- System-generated timestamps
- Different processing times
- Not true “errors”

3- **Preliminary Data Issues:**

- Completely Empty Column
- Data type risk
- Column naming

4- Exploratory Data Analysis (EDA) Report

This report presents an initial exploration of the dataset using missing value Table insight, Bar Chart visualization, visualizations, summary statistics, and pattern detection.

Dataset Snapshot

- Total Records: 7,544
- Total Variables: 80
- Most variables are categorical, while a smaller subset are numeric (primarily IDs and timestamps).

A- Missing Values : Table Insight

From the dataset (7,544 rows, 80 columns), the table shows that:

- Several columns have very high missing values (over 90% missing).
- Examples include:
    1. Most_Recent_Released_Decision
    2. SLU_Start_Comment
    3. RIT_Email_Created
    4. Housing_Contract
    5. Is_Global_Grad

Interpretation:

These columns are sparsely populated and may:

- Represent optional processes
- Be system-generated fields
- Not be useful for core analysis

B- Visualizations

1- Missing Values : (Bar Chart)

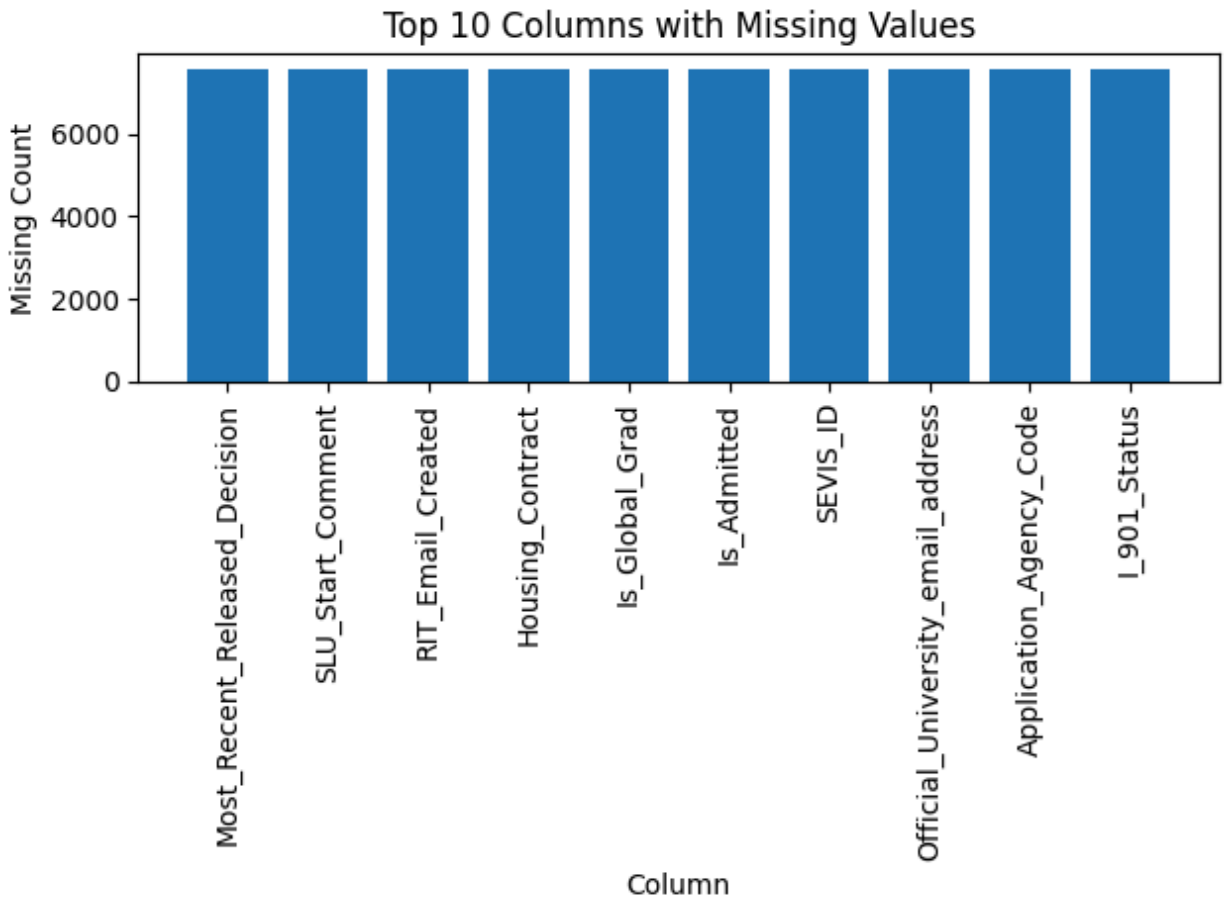


Chart: “Top 10 Columns with Missing Values”

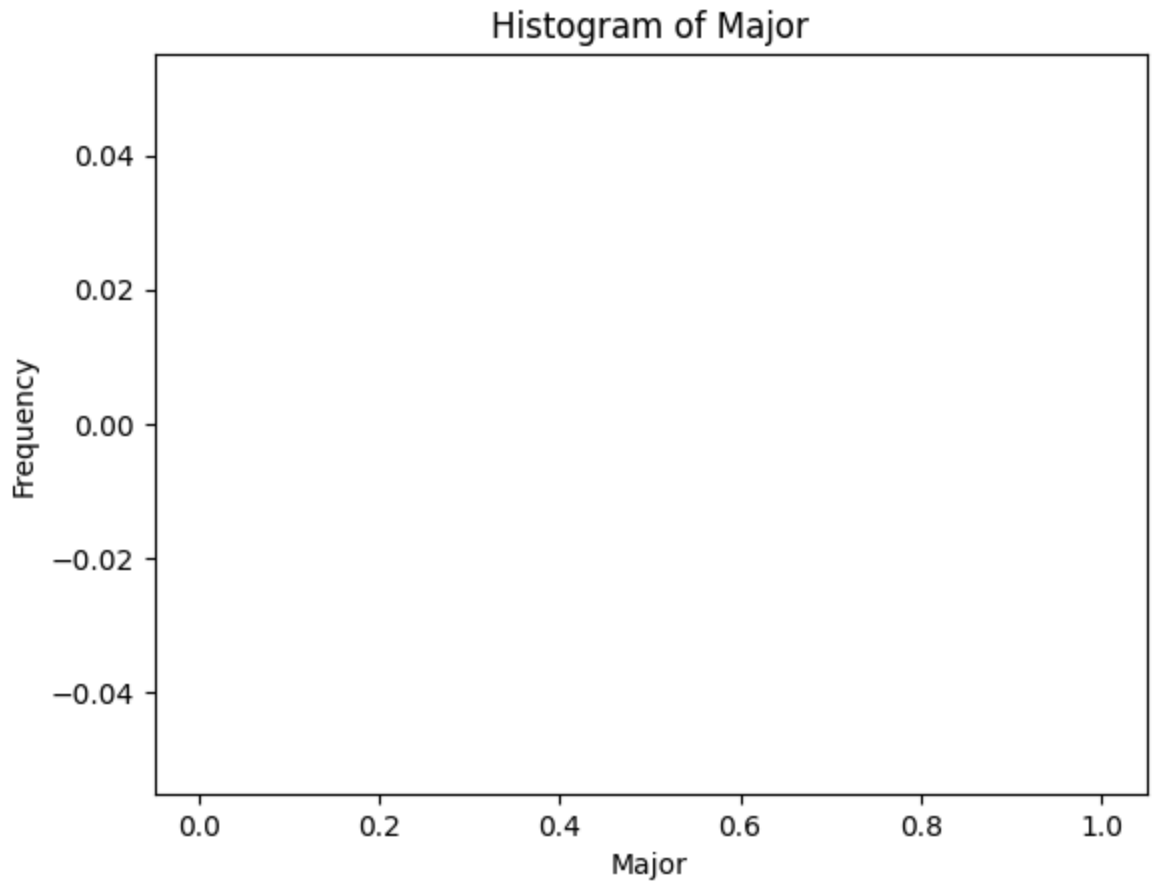
What the chart tells us:

- The bars are almost equal in height, showing that many of these fields are nearly empty across the dataset
- This visually confirms the data completeness issue in these columns

Why this matters:

- High missingness reduces analytical value
- Including these columns may bias results or reduce usable data size.

2- Histogram

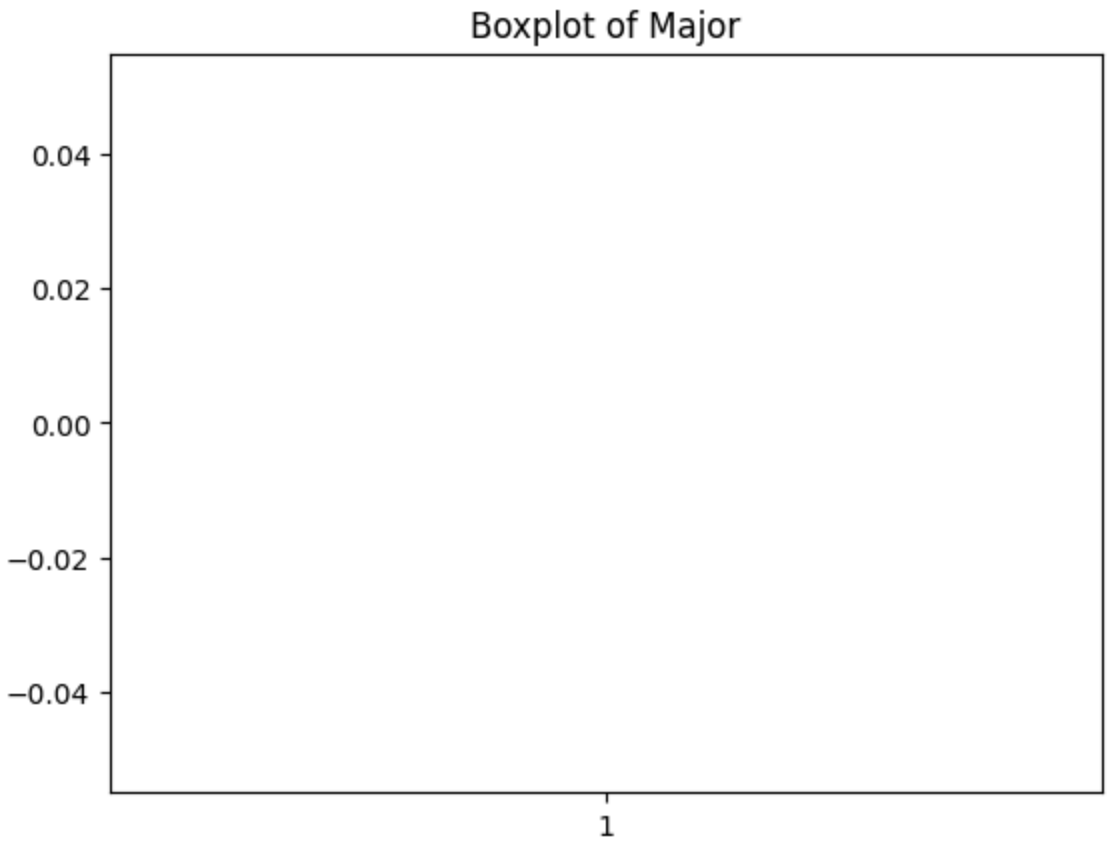


Histogram of “Major”

Observation:

The chart appears empty because the Major column has no numeric data — confirming it is completely missing and unsuitable for analysis.

3- Boxplot:



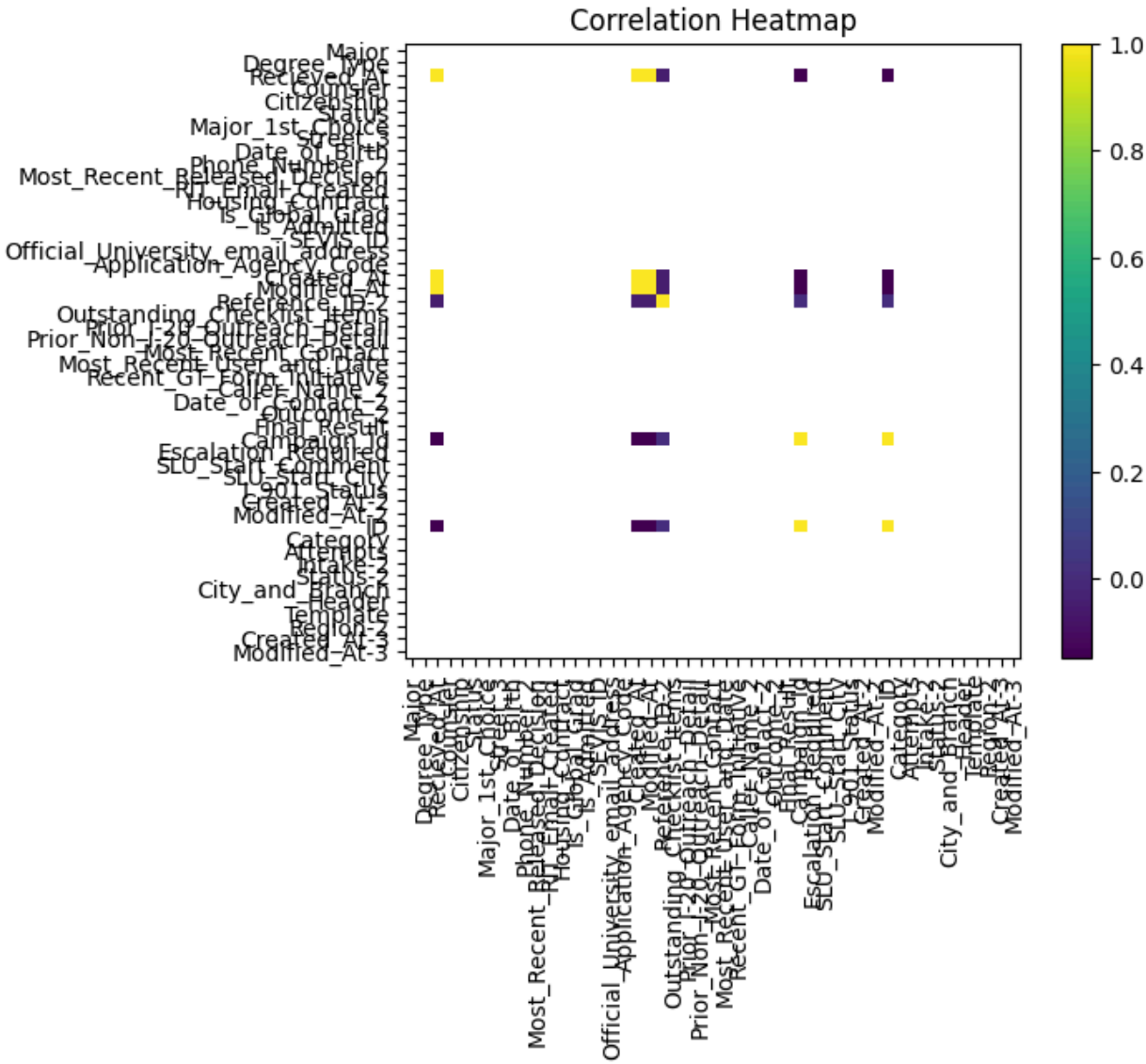
Boxplot of “Major”

Observation:

No distribution exists due to missing values.

Insight: We need to reinforce the need for column removal.

4- Correlation Heatmap:



What it shows:

- Very few strong correlations between numeric fields.
- Most numeric columns behave independently.

Important Note:

Many numeric columns are actually:

- System IDs
- Campaign IDs

- Timestamp values

These cannot be used for correlation analysis, as the relationships are not meaningful.

C- Summary Statistics

From the computed statistics:

1. Timestamp Fields

Examples:

- Received_At
- Created_At
- Modified_At

Characteristics:

- Very small standard deviation
- Nearly identical min, median, and max

Insight:

From findings, this data was likely generated within a narrow timeframe - possibly a bulk import or synchronized system update.

2. ID-Based Columns

Examples:

- Campaign_Id
- Reference_ID
- ID

Observation:

- Wide numeric ranges

- Large standard deviations

These are identifiers, therefore not measurable quantities.

Not treated as continuous variables.

D- Early Patterns & Distributions

1. High Column Sparsity:

Many variables contain little to no data.

Suggests:

- Optional fields
- Process-driven entries
- Export artifacts

Risk: Too many sparse columns reduce model performance.

2. Structured System Data:

Large timestamp and ID presence indicates the dataset likely comes from:

CRM / admissions workflow software

This is typical in operational datasets.

3. Low True Numeric Content:

There are very few genuine quantitative variables.

Analysis will rely heavily on:

- Categorical grouping
- Frequency analysis
- Status tracking

rather than regression-style modeling.

E- Key Insights:

Strengths:

- Large dataset
- Structured format
- Minimal duplicates

Risks:

- Completely empty columns
- Many system-generated numeric fields
- High sparsity in several variables

This report presents an initial exploration of the dataset using missing value, visualizations, summary statistics, and pattern detection.